Final Exam, Advanced Statistics and Alternative Data Types

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December 5th 2025

Question 1

Question 2

(i)

Question 3

(i) Below is our plot of the data

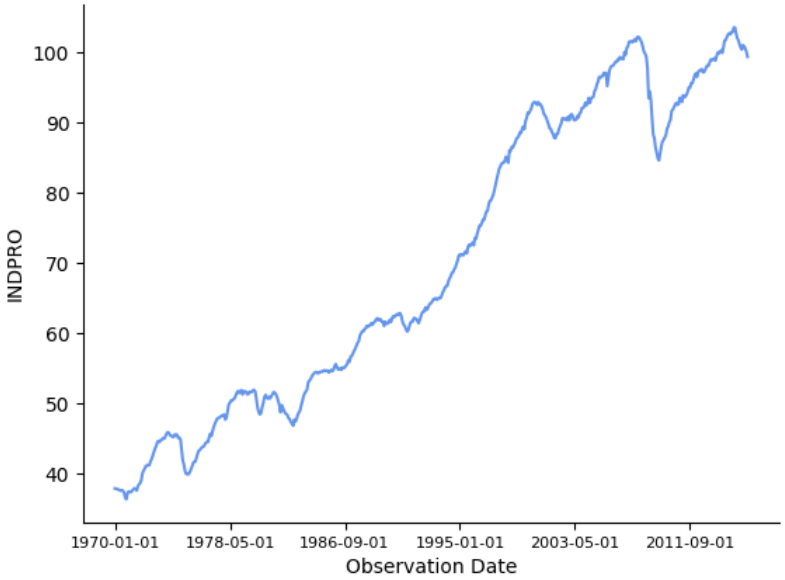


Figure 1: Industrial Production from 1970 to 2015

(ii) The dickey-fuller unit root test gave a p-value of around 0.748. Which was not very surprising, considering the figure in (i) displays a very non-stationary time-series. W

(iii) Using the approach in the notes [2], p. 149. We know that if we difference the series enough we can make it stationary. Using the methodology from the **ADFuller.ipynb** we will difference the series until we get a satisfactory p-value. We made an arbitrary choice of setting our threshold ex-ante to 5-percent. The result gave us a ARIMA$(p, 1, q)$

(iv) For this question we use the methodology from **ts_estimation.pdf**. As in the slides we denote that ρ as the ACF and α as the PCAF. The sample α (figure 1) and ρ (figure 2) both show exponential decay which suggest that this is an ARMA process. We use a decision rule of selecting order based upon the last lag outside the 5-percent band. And therefore choose an ARMA $(7, 5)$.

(v) We used the code from **ARMA(1).ipynb** but modifed it to use AICC instead of AIC. Our minimial AICC model was ARMA$(4, 0)$ and our minimal BIC model was ARMA$(1, 1)$. We went with the BIC as

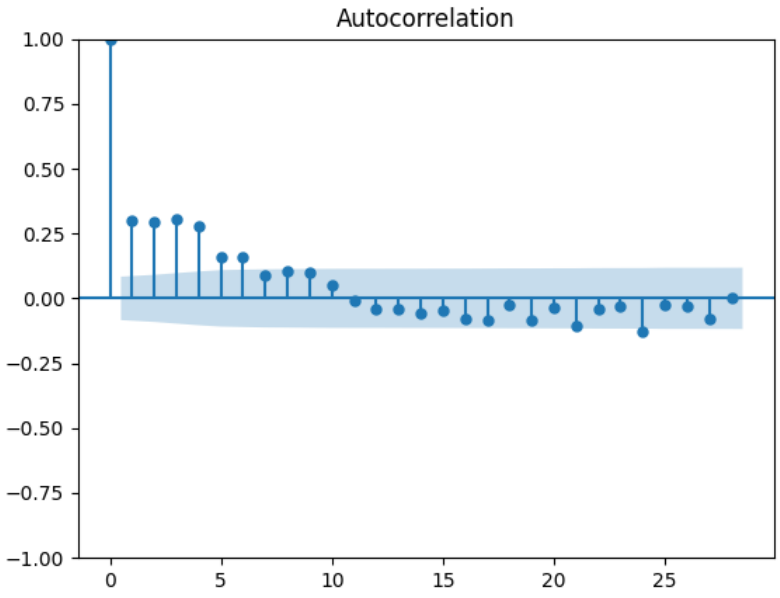


Figure 2

it aligns closer with our interpretation of with our sample ρ plot. As we suspected some MA component. It also involved fewer lag orders.

(vi) Based on the results in (v) we estimate a model of the process

$$X_t = \phi X_{t-1} + \theta \epsilon_{t-1} + \epsilon_t \tag{1}$$

Firstly, we chose again a significance level of 5-percent. The model results showed that both AR and MA components were statistically significant. Our estimate of the AR(1) component $\hat{\phi}$ was around 0.87, which is quite high. This implies that past shocks decay slowly. Furthermore, our estimate of the MA(1) component $\hat{\theta}$ was around -0.64. The decently high negative sign implies that a shock in the previous period is offset to some degree in the next period.

(vii) Below is the plot of our forecast

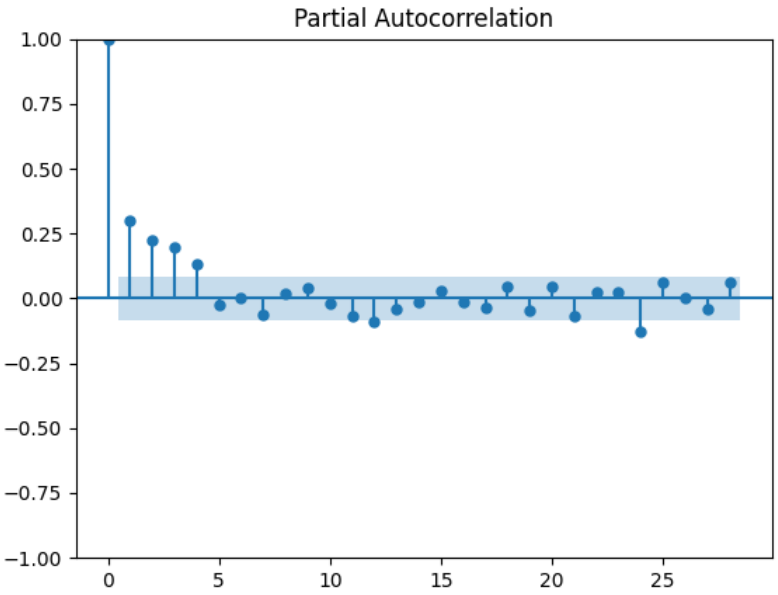


Figure 3

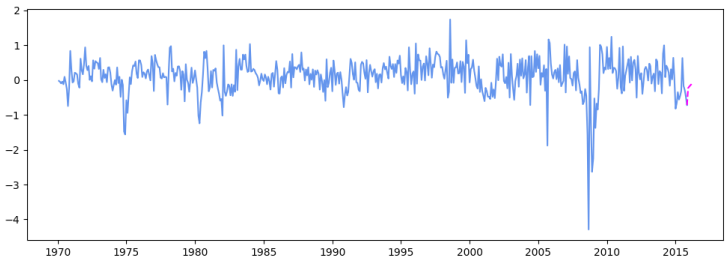


Figure 4

Bibliography:

References

[1] Casella, George, and Roger L. Berger. *Statistical Inference*. 3rd ed. Boca Raton, FL: CRC Press, 2024.

[2] Lee, Adam. *Notes*. BI Norwegian Business School, 2025.